What determines the success of a game?

Antonio Contestí Coll, Josep Ferriol Font,
Francesc Gayà Piña, Marc Link Cladera,
Carlos Gálvez Mena, Jordi Vanyó Mestre,

*Abstract—This study explores the determinants of video game success on the Steam platform [1], [2]. **The primary challenge of this research lay in the preprocessing phase**: the original dataset exhibited severe quality issues, massive inconsistencies, and biases that required an exhaustive and complex cleaning process to render useful information. Upon establishing a reliable data foundation, we employed multivariate **clustering** (GMM) to segment the market, identifying a clear "success" profile. Our analysis of this profile reveals that a game's potential is encoded in its **"DNA"**—defined by structural variables such as global scalability (languages) and mechanical depth—and amplified by its **economic accessibility** (Price). We conclude that success is a structurally identifiable phenomenon, dependent on measurable inherent characteristics rather than just semantic descriptions.*

The video game industry has experienced exponential growth over the last decade, cementing Steam as the leading digital distribution platform for the PC market. With thousands of titles released annually, understanding the mechanisms that distinguish a commercial success from a failure has become a complex issue requiring a data-driven approach.

However, any rigorous analysis is intrinsically dependent on the quality of the underlying information. This is where the primary challenge of this research phase lies: a cornerstone of this project has been the rigorous collection, cleaning, and validation of data. The original dataset exhibited severe shortcomings, as the data derived from previous scraping efforts **showed significant discrepancies with the reality of the Steam platform**. These inconsistencies were particularly acute in key variables such as review volume, sentiment analysis, and external ratings (Metacritic [3]).

Due to the unreliability of the original source, we decided to conduct our own data extraction using the official Steam API. This strategic decision allowed us to rectify the errors and correct the formatting of numerous features that were originally unsuitable for robust statistical analysis. Consequently, this document focuses on the process of building a reliable data infrastructure, an indispensable prerequisite for the subsequent modeling and extraction of insights regarding success in video games.

Following this initial preprocessing phase, we categorized the dataset variables into two distinct groups: **feature variables** and **performance variables**. Performance variables are defined as the game's output metrics—those dependent on its intrinsic characteristics—such as the number of owners, review count, global rating, user sentiment, and total playtime. These metrics enable us to quantify a game's market performance.

Subsequently, we performed a cluster analysis utilizing these performance variables. This approach grouped games exhibiting similar market behaviors into the same cluster, effectively enabling us to segregate successful titles from underperforming ones.

Finally, leveraging this grouping, we analyzed the disparities in feature variables across the different clusters. This comparative analysis aims to isolate and identify the specific characteristics that define a successful video game.

1. Dataset overview

The original dataset comprises a total of 89,618 observations and 47 columns. We conducted a systematic analysis of each variable to comprehend the information provided and to determine its relevance to our study.

Beyond the technical data types, it is essential to categorize the variables based on their semantic nature and the specific dimension of the video game

market they represent. The raw dataset offers a multidimensional view of each title, which can be conceptually structured into five main analytical domains:

- **Intrinsic game characteristics**: These variables define the identity and content of the product. This group encompasses descriptive metadata such as `genres`, `categories`, `notes`, and `tags`, which classify the game's style and mechanics. Additionally, description columns such as `about_the_game` and `detailed_description` provide context regarding the game's features and gameplay.
- **Technical characteristics**: These features represent the intrinsic non-in-game characteristics of the product. This includes platform compatibility (`windows`, `mac`, `linux`), content volume indicators such as `dlc_count` and `achievements`, author attribution information such as `developer` and `publisher`, and language information with `supported_languages` column.
- **Commercial and economic metrics**: This category includes variables related to the monetization and market reach of the title. Key variables here are `price` (the barrier to entry), `discount`, and `estimated_owners`, which serves as a proxy for sales volume. Additionally, `release_date` provides the temporal context necessary to adjust for market trends.
- **Player engagement and retention**: To understand how players interact with the game, the dataset provides telemetry data regarding time investment. Variables such as `average_playtime_forever` and `median_playtime_forever` offer insights into retention and replayability, while `peak_ccu` (Peak Concurrent Users) indicates the game's maximum popularity at a specific point in time.
- **Community reception and feedback**: Finally, the dataset captures the subjective quality of the game through user and critic feedback. This includes the raw volume of feedback (`reviews`, `recommendations`, `num_reviews_total`), the sentiment balance (`positive` vs. `negative` ratings), and external validation via the `metacritic_score`. These metrics are critical for assessing the "reputation" of a title within the community.

However, a subset of the variables does not fit neatly into these analytical categories. This residual group primarily consists of **auxiliary and administrative metadata**, such as direct URLs (e.g., `support_url`, `metacritic_url`), visual assets (e.g., `header_image`, `screenshots`), and internal identifiers. For these cases, we adopted distinct handling strategies depending on the nature of the information: while some were discarded as noise, others were flagged for feature extraction in the subsequent preprocessing phase.

2. Data engineering

2.1. Sample definition and noise reduction

The first step consisted of an exhaustive cleaning of the original data. Since the main objective of the project is to identify the characteristics that define success in video games, it was necessary to discard those records that do not provide relevant information for this analysis.

An exclusion heuristic was established based on user interaction: titles with a **median playtime of less than 30 minutes** that, simultaneously, **do not have any reviews** were removed. The application of this criterion reduced the data volume by **90%**, going from more than 80,000 initial entries to a final sample of 6,500 records.

Although this reduction is drastic, it is essential to eliminate the statistical noise generated by games with no activity or *feedback*.

The decision to set the threshold at 30 minutes is based on the nature of certain genres: there are historically successful and high-impact games (such as the archetype of *Snake* or *Minesweeper*) designed for short or casual sessions. Excluding games based solely on high playtime would have eliminated relevant titles from this niche.

2.2. Quality audit and handling of inconsistencies

Once the sample of 6,500 records was defined, a consistency study was carried out on the variables that presented anomalies during the initial exploration. The analyzed columns include, among others: `metacritic_score`, `positive`, `negative`, `reviews`, and derived data such as `pct_pos_total` or `estimated_owners`.

The following critical inconsistencies were detected:

- **Missing Data**: The Metacritic rating variable (`metacritic_score`) presented a large amount of null values, even though the score is publicly accessible.
- **Arithmetic Incoherence**: In numerous cases, the sum of positive and negative reviews did not match the value of `num_reviews_total`.

Consequently, the derived percentages (`pct_pos_total`) were erroneous.

Given the lack of reliability of the original data regarding reviews, we opted to discard these columns and perform a *data enrichment* process using *scraping* techniques.

2.2.1. External Data Acquisition and Sentiment Analysis
To address the detected deficiencies, a data extraction module was implemented using the official Steam API [4] and a script to extract data from the SteamDB website [5]. This process allowed us to obtain granular and reliable information divided into two levels:

Review Level:
For each individual comment, we extracted:

- **The language of the review**.
- **The associated sentiment analysis** (whether it is positive or negative).

We discarded the text of all reviews because the sentiment analysis is already provided by the Steam API itself.

In this way, synthesizing, we added four additional columns per language to each game: **quantity of positive reviews**, **quantity of negative reviews**, **quantity of positive reviews from the last two weeks**, and **quantity of negative reviews from the last two weeks**. This generated a total of **120 columns**.

Game Level (aggregated data):
The global metrics were updated:

- **Review score and its textual description**.
- **Exact count of positive, negative, and total reviews**.
- **Historical peak of concurrent players.**

From this new extraction, two differentiated *datasets* were generated: **one with global performance metrics**, which will be used for the initial clustering, and **another with the detail of the reviews**, intended for the qualitative analysis of the resulting clusters.

The resulting global metrics dataset consists of the variables described in Table 1.

The column `all_time_peak` (historical peak of concurrent players), obtained via SteamDB scraping [5], will serve as an additional metric to evaluate the success of a title based on its maximum volume of users.

TABLE 1. Variables of the global metrics dataset

Variable	Description
`appid`	Unique identifier of the game on the Steam platform.
`review_score`	Numeric score associated with the global rating extracted from the API.
`review_score_desc`	Textual label of the rating category (e.g., *Very Positive*).
`total_positive`	Total count of accumulated positive votes.
`total_negative`	Total count of accumulated negative votes.
`total_reviews`	Total sum of all reviews (positive and negative) registered.

2.3. Metadata structuring and categorical variables

First, we focused on **categorisable text variables** such as `notes`, `tags`, `categories`, and `genres`. We considered them categorisable because their potential dimensions are structurally limited, since we are not dealing with free or extensive natural language text.

The `notes` field contains sensitive content warnings, such as violence, sexual content, or substance use. This attribute is optional and is only filled in when a game includes elements that require a specific warning. Therefore, the presence of null values is inherent to its design and is not considered missing information. It has a highly repetitive structure, based on standardized phrases and a limited vocabulary. Given this semi-structured nature, we opted for a rule-based approach relying on keyword detection (discarding the use of advanced natural language processing techniques). We applied basic preprocessing to the text, including the removal of HTML tags and stop words, as well as lowercasing. Afterward, we defined semantic groups representing the main types of content warnings, and for each group we generated a binary variable indicating the presence or absence of that content in the game. This procedure allowed us to transform a heterogeneous textual field into a set of structured, interpretable variables that can be easily integrated into subsequent analyses. The following list shows the resulting binary variables:

1) **Violence**
2) **Gore**
3) **Nudity**
4) **Substance Use**
5) **Offensive Language**
6) **Horror**
7) **Self-Harm**
8) **Adult Content**
9) **Gambling**

Each of these categories becomes a column where every game takes the value `TRUE` or `FALSE` depending on whether it contains that type of content.

Regarding the `tags` field, we extracted an initial numeric metric that computes **how many times a tag has been assigned to a game**. We included this value because we considered it a **performance-related metric** that could help in the clustering process, since it depends on user interaction with the game. In addition, we cleaned the field so that only the tag labels remained, removing the associated counts. This will allow us to more easily study the distribution of tags once the resulting groups are available.

For the `categories` column, we performed a manual filtering step. Since there were only 40 categories, we removed those we considered less informative, namely: “Commentary available”, “Steam Turn Notifications”, “Includes Source SDK”, “SteamVR Collectibles”, and “Steam Timeline”. We also merged the categories “VR Supported” and “VR Support”.

Finally, we transformed the `genres` variable using one-hot encoding, and removed values that overlapped with the `notes` field, namely: “Sexual Content”, “Nudity”, “Gore”, and “Violence”. This resulted in a total of **18 different genres**.

2.4. Natural language processing (NLP)

The original dataset includes several unstructured textual variables, specifically `short_description`, `about_the_game`, `detailed_description`, and `reviews`. These columns contain descriptive information about the video game, mainly generated by developers, as well as user opinions in the case of `reviews`.

First, we decided to exclude the `reviews` variable from this textual analysis block. This decision is based on methodological criteria: `reviews` constitute an *ex post* source of information, strongly influenced by the game’s real performance, and they are analyzed separately in a later section using a dataset obtained via *web scraping*. In this way, we avoid mixing pre-release information with indicators that directly reflect the market response.

Regarding the descriptions provided by developers, a preliminary exploratory analysis revealed a high degree of redundancy among the `short_description`, `about_the_game`, and `detailed_description` columns. In most cases, these three variables contain semantically very similar information, with differences mainly related to text length and the level of detail.

For this reason, and in order to reduce model complexity and avoid an artificial duplication of textual signal, we chose to work only with the `about_the_game` variable. This column generally presents a richer and more descriptive vocabulary, with a higher presence of terms related to gameplay mechanics, setting, and user experience, making it an appropriate representation of the product’s semantic content.

As a first step in the exploratory analysis, we examined the distribution of the most frequent words in this column after a basic text-cleaning process (lowercasing, removal of punctuation, and removal of stop words). Figure 1 shows the twenty most common words, confirming the use of relatively homogeneous promotional and descriptive language across video games.

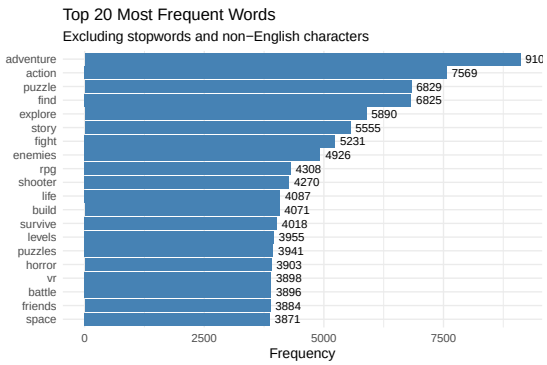

FIGURE 1. Top 20 most frequent words in the `about_the_game` column.

Once the suitability of this column was validated, the text was transformed into a numeric representation using the **TF-IDF** (*Term Frequency–Inverse Document Frequency*) method. This technique allows the relative importance of each term within the corpus to be weighted, reducing the impact of very frequent but poorly informative words and highlighting more discriminative terms.

Given the high dimensionality of the resulting TF-IDF matrix, several aggregated metrics of the vector were computed (norm, number of non-zero terms, maximum value, and mean), with the aim of obtaining quantitative variables that summarize lexical richness and text structure without relying on the full vocabulary.

In parallel, a dimensionality reduction technique based on **Singular Value Decomposition (SVD)** was applied, obtaining a reduced set of latent components that capture the main semantic patterns of the text. These components represent abstract dimensions of textual content, related to communicative styles, broad themes, or product presentation strategies, and they enable semantic information to be incorporated into subsequent models at a reduced computational cost.

The components obtained through SVD are later used both in exploratory analysis and in the *clustering*

and classification processes, either independently or combined with the video game's structural variables. This approach makes it possible to assess the extent to which textual content contributes to explaining performance differences or market trajectories, while maintaining reasonable model interpretability.

Overall, this preprocessing and exploratory natural language analysis process allows textual information to be integrated into the project's general data mining workflow, highlighting both its descriptive potential and its limitations when explaining complex phenomena such as the commercial success of video games on Steam.

2.4.1. Determining the optimal number of clusters
Once the vector representation of textual content was obtained through TF-IDF and dimensionality reduction was applied using SVD, semantic segmentation of video games was performed using *k-means*. A critical aspect of this process is determining the optimal number of clusters, since this parameter conditions both the quality of the segmentation and its subsequent interpretability.

To address this decision systematically, two standard criteria in unsupervised analysis were used: the **Elbow Method** and the **Silhouette Score**.

First, the analysis using the Elbow Method did not reveal a clear inflection point. The within-cluster inertia decreases progressively as the number of clusters increases, without the emergence of a clear "elbow" indicating an optimal value of k. This behavior suggests a latent textual structure with diffuse boundaries, which is common in high-variability semantic data such as video game descriptions.

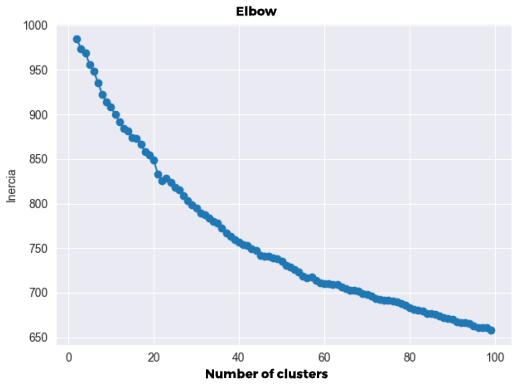

FIGURE 2. Elbow Method applied to *clustering*. Inertia decreases progressively without a clear inflection point, which is common in semantic data with diffuse boundaries.

Complementarily, the silhouette coefficient was computed for different values of k. The results show that the maximum *silhouette score* is achieved at $k = 3$, indicating that this solution provides the best average separation between clusters. However, beyond this value a relatively stable **plateau** is observed for higher values of k, with slightly lower scores but without an abrupt degradation in quality.

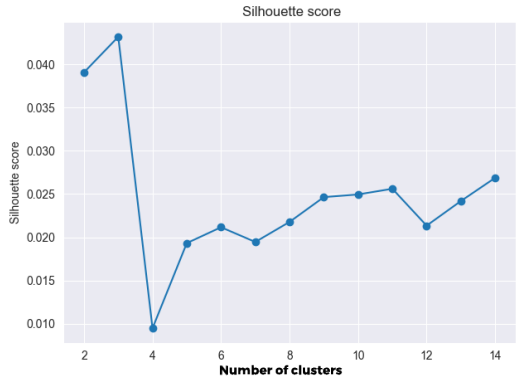

FIGURE 3. Silhouette coefficient as a function of the number of clusters (k). A maximum is observed at $k = 3$, followed by a plateau for higher values, indicating several reasonable *clustering* solutions in terms of quality.

This behavior suggests that, although a three-cluster solution is optimal from a strictly geometric standpoint, solutions with a larger number of clusters still maintain reasonable quality and allow a finer segmentation of the semantic content.

Taking this trade-off between quality and interpretability into account, we ultimately selected a solution with $k = 7$ **clusters**. This choice makes it possible to capture different themes and game styles in greater detail, facilitating a clear semantic interpretation of the resulting groups, without significantly penalizing internal cohesion.

It is important to note that these seven clusters are **semantic** in nature and depend exclusively on the textual content of the video game descriptions. Therefore, they do not coincide with the seven performance segments defined in other parts of the project, which are based on quantitative adoption and commercial success metrics.

Based on a qualitative analysis of the predominant terms in each cluster, the following interpretable labels were assigned:

- **Cluster 0**: Realistic simulation and management
- **Cluster 1**: Open world, exploration, and survival
- **Cluster 2**: Puzzles and casual experiences
- **Cluster 3**: Multiplayer and competitive action
- **Cluster 4**: Tactical strategy and turn-based RPG

- **Cluster 5**: General action and combat
- **Cluster 6**: Narrative adventure and exploration
- **Cluster 7**: General adventure experiences

This nomenclature helps connect the *clustering* results with recognizable conceptual categories in the video game market, facilitating both exploratory analysis and the interpretation of results obtained in subsequent prediction stages.

2.5. Entity resolution

Within the publishers and developers columns, we encountered numerous distinct values that, in reality, refer to the same entity. For example, "Ubisoft Montreal" and "Ubisoft Quebec" are the same entity, Ubisoft, but with different locations. In other cases, we found that many companies begin with generic names or stopwords: "Black", "Red", "The", and similar terms.

To eliminate this "duplicity", we opted for a brute-force solution. We merged values sharing a common prefix while disregarding these stopwords, to avoid oversimplifying all entities that begin, for instance, with "The" by assigning "The" as the common entity.

This process is divided into two parts:

- Dynamically generating a map that directly associates the list of games with similar names to the common part.
- Cleaning the names using information from the map, substituting the full names with the value associated in the map.

2.6. Characterization of monetization strategy

To complete the variable transformation, the `packages'` column was processed. This field contains, in JSON format, a mixed list of all purchasable items related to the game. This includes everything from different editions of the base title to microtransactions (such as virtual currency or expansion packs) and subscription models.

Since the original structure did not allow for direct analysis, we separated these elements into three functional categories:

- **Versions:** Game editions (standard, *deluxe*, etc.).
- **Subscriptions:** Recurring payments.
- **Micropayments:** Purchases of additional content or in-game currency.

Based on this classification, we generated nine numeric variables that summarize the commercial offering of each game, capturing both the quantity of options and the price range (minimum and maximum) for each of the three categories 2.

TABLE 2. Variables generated for economic analysis

Category	Generated Variables

Versions	`versions_count`
	`min_version_price`
	`max_version_price`

Subscriptions	`subscriptions_count`
	`min_monthly_sub_price`
	`max_monthly_sub_price`

Micropayments	`micropayments_count`
	`min_micropayment_price`
	`max_micropayment_price`

This separation allows us to distinguish between single-payment games and those that base their model on microtransactions or subscriptions.

2.7. Quantification of resources and links

Finally, variables containing external references in URL format were treated. The text of these links does not provide analytical information per se, but their existence or quantity are useful indicators of the product sheet's completeness and marketing effort.

To integrate this information, we applied two transformation strategies depending on the case:

- **Presence indicators (Binary variables)**: In cases where the only relevant factor is knowing whether the resource exists, we replaced the URL with a boolean variable. For example, the `website` column was transformed into `has_website` (with a value of `TRUE` if the original field was not null).
- **Resource count (Numeric variables)**: In fields containing lists of multiple URLs, such as images or videos, we opted to count the quantity of elements. An example is the `screenshots'` column, which was converted into `screenshots_count`, thereby capturing the amount of graphic material available to the user.

2.8. Transformation and derivation of new variables

Beyond the group-specific cleaning procedures, a general transformation pipeline was applied to standardize key variables and generate new ones to facilitate subsequent analysis.

2.8.1. Owner estimation The variable `estimated_owners` was originally presented

as a textual range (e.g., “20000 - 50000”). To utilize this information in numerical models, we calculated the arithmetic mean, creating the new variable `owners_avg`. This simplification converts an uncertain interval into a robust point estimator.

2.8.2. Temporal variables The release date (`release_date`) was decomposed to capture seasonality patterns: the year, month, and day of the week were extracted. Even more significant is the creation of the variable `days_since_release`, calculated as the difference between a fixed cutoff date (01/04/2025) and the release date. This metric is fundamental, as it will act as a denominator to normalize cumulative variables (such as total reviews or votes), enabling a comparison on equal footing between newly released games and veteran titles (daily rate vs. accumulated total). We thus observe that it also plays a pivotal role in a different manner.

2.8.3. Logical imputation and list cleaning Finally, structural inconsistencies were corrected:

- **Sentinel values**: Values of `-1` in fields such as required age or percentages were converted to `0` to maintain numerical consistency.
- **Empty publishers**: For games with no reported `publisher`, a self-publishing model was assumed, imputing the value from the `developer` column.
- **Lists**: Text strings representing lists (publishers, developers, genres) were cleaned, removing superfluous characters (such as brackets or quotation marks) and standardizing the separator.

2.9. Final attribute selection

Once we have the filtered data available, we must also select which columns to retain and which to discard for our analysis.

Firstly, we disposed of those presenting inconsistencies (the ones previously mentioned), as we obtained them anew via scraping.

Subsequently, we eliminated those that did not contribute relevant information, such as `header_image` and `metacritic_url`.

3. Clustering — First approach

To identify what determines the success of a video game, we will first define success groups based on variables derived from user interaction (these variables will be collected under the name "performance variables"). This information includes both, direct feedback (reviews and recommendations) and indirect feedback (playtime or simultaneous players). Specifically, the following variables have been analyzed:

- `median_playtime_forever`: Median time played per player.
- `median_playtime_2weeks`: Median time played in the last two weeks per player.
- `recommendations`: Number of times the game has been recommended.
- `owners_avg`: Approximate number of owners.
- `total_positive`: Number of positive comments.
- `total_reviews`: Number of comments.
- `num_tag_association`: Number of times players have assigned a tag to it.
- `all_time_peak`: Maximum number of concurrent players.

3.1. Specific preprocessing

3.1.1. Outliers Prior to the clustering process, data cleaning was performed to eliminate biases due to the use of bots. Titles with anomalous playtimes (exceeding 17h/day) were detected, generally linked to card farming for sale on the Steam market. To filter these cases, a K-Means algorithm ($K = 2$) was applied, managing to separate normal values from abnormal. As illustrated in Figure 4, the large distance between both distributions allows for a clear classification of these outliers.

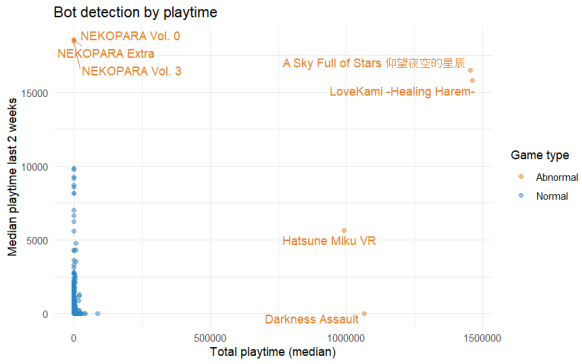

FIGURE 4. Playtime outliers.

3.1.2. Correcting distributions Given that most variables show a right-skewed long-tail distribution, the most popular games distort distances due to their large magnitude. By applying a logarithmic transformation, we managed to "compress" the range of the highest values and expand the low ones, facilitating a clearer

differentiation between medium-success games. Figure 5 shows how the variables are better distributed after this process.

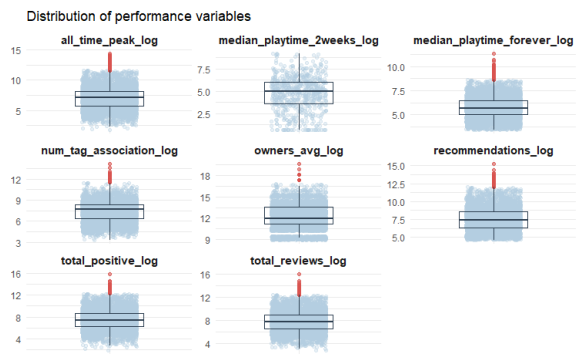

FIGURE 5. Box plots of performance variables.

Furthermore, with the aim of comparing games on equal terms, we have converted cumulative performance variables into daily ratios (dividing by `days_since_release`). This normalization, together with the logarithmic transformation, allows us to capture the relative success of each game regardless of its time since release, facilitating the identification of success patterns in both recent releases and consolidated games.

3.1.3. Correlation analysis Before applying clustering techniques, we analyzed the correlation between the selected performance variables with the aim of mitigating multicollinearity. The analysis reveals two clearly differentiated groups of variables (see Figure 6):

- **Independent variables**: Metrics for median playtime and peak simultaneous players have a low correlation, indicating that they capture different dimensions of user behavior.
- **Redundant variables**: A near-perfect correlation ($r \geq 0.95$) is observed between review volume, recommendations, owner estimation, and tag associations. This redundancy suggests that these metrics describe the same phenomenon: the scale of the title's daily growth.

3.1.4. Consolidation using PCA To resolve the detected multicollinearity, we applied a Principal Component Analysis (PCA) on the block of growth variables. With a single principal component (PC1), we are able to explain 97.75% of the total variance, while the second component only adds an additional 1.17%.

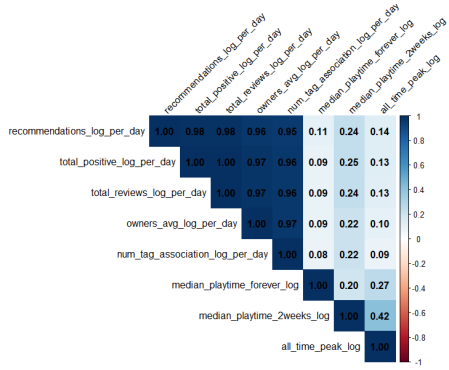

FIGURE 6. Performance variable correlation matrix.

Given the low contribution of this second dimension and following Kaiser's criterion [6], which suggests discarding components with an *eigenvalue* below 1 (PC2 $\approx$ 0.24, see Figure 3.1.4), we have opted to retain only PC1. Because all original variables load positively and equally on this component, the component has been renamed as `growth_log_rate`, consolidating it as a unified metric of success and adoption.

	PC1	PC2	PC3	PC4	PC5
Std. dev.	2.2110	0.2362	0.1802	0.1462	0.0189
Prop. var.	0.9780	0.0112	0.0065	0.0043	0.0001
Cum. prop.	0.9780	0.9892	0.9957	0.9999	1.0000

TABLE 3. Importance of components (PCA).

3.1.5. Resulting final variables After all this preprocessing, subsequent clustering models will be based on four decorrelated variables that cover the characteristics we find essential to determine success:

- Retention: `median_playtime_forever`
- Recent engagement: `median_playtime_2weeks`
- Historical reach: `all_time_peak_log`
- Popularity and growth: `growth_log_rate` (PC1)

This simplification allows us to use Euclidean distance, avoiding the bias that redundant variables would produce and eliminating the need to resort to more complex metrics such as Mahalanobis distance.

3.2. Evaluation of clustering techniques

3.2.1. Grouping tendency and initial exploration Prior to applying the algorithms, we evaluated the clustering tendency of the dataset using the Hopkins test. The

result of 0.9467 indicates a very high tendency of the data to form non-random clusters. An initial inspection using 3D plots with retention, *engagement*, and popularity values allowed us to observe that the resulting clusters could be fuzzy.

3.2.2. Limitations of density-based methods (DBSCAN) Initially, we tested DBSCAN with the objective of identifying groups by point density. However, the results were not satisfactory due to the distribution of the `median_playtime_2weeks` variable. There is a large concentration of games with a value of 0 (titles without recent activity), which generates a high-density zone that "traps" most records into a single cluster, while the rest remain dispersed as noise or in residual groups. Even after filtering the sample to include only active games (`median_playtime_2weeks` > 0), DBSCAN failed to segment the population in a revealing way with the variables introduced in the model.

3.2.3. Hierarchical clustering and imbalance After obtaining no significant results through densities, we opted for algorithms based on distances to centroids. Hierarchical clustering was applied using the *average link* method to games that had been active in the last two weeks. The analysis of the dendrogram (see Figure 7) does not show a clear cut-off structure. When forcing a solution of $k = 4$ groups to explore qualitative differences, we obtained one massive cluster compared to three minority groups. This critical imbalance prevents drawing statistically significant conclusions about the behavior of the small groups. For this reason, we decided not to go deeper into this path and to try a different preprocessing, leaving this approach as a first approximation.

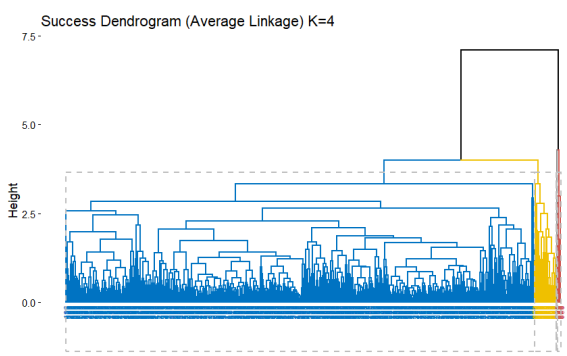

FIGURE 7. Dendrogram of hierarchical clustering on active games.

4. Clustering — Second approach

4.1. Preprocessing

4.1.1. Normalization discrepancies After analyzing the results obtained in the first approach, we detected a very important fact. In the second stage of the first approach, we normalized the cumulative variables by dividing them by the days since release (`days_since_release`). However, this way of handling the metrics was heavily biasing the results! Variables that initially showed no significant correlation suddenly exhibited nearly a 100% correlation simply due to being divided by the days! This behavior was neither coherent nor desirable; thus, after a deeper analysis, we concluded that having a **common denominator** was the root of the problem.

If we have two variables A (reviews) and B (owners), it is possible that they share a certain relationship, but not a perfect one. If we introduce a third variable T (days) and create $X = \frac{A}{T}$ and $Y = \frac{B}{T}$, if the variance of T is significantly greater than that of A or B, then T begins to dominate the equation:

1) **New games (small T)**: Both X and Y will be very large numbers.
2) **Old games (large T)**: Both X and Y will be very small numbers.

Essentially, we are ordering the games based on their age. The variables move together not due to a genuine synchronization between reviews and owners, but because **both are affected with the same intensity by the denominator T**.

Based on this insight, we decided not to divide the cumulative variables by the number of days since release, but rather to use `days_since_release` as an additional explanatory variable! As can be observed in Figure 8, the correlation between the cumulative variables is significantly lower! Furthermore, the correlation with the days since release is negative!

4.1.2. Consolidation via PCA At this stage, we aimed to proceed in the same manner as in the first approach; that is, grouping variables with high inter-correlation into a single column using PCA. The selected variables are: `recommendations_log`, `owners_avg_log`, `total_positive_log`, and `total_reviews_log`, as they all share at least a 75% correlation with each other.

To do this, we first wanted to ensure that the variables possess a linear relationship. As seen in Figure 9, we can assume linearity and perform PCA.

The first component (PC1) explains a large portion of the variance, specifically **81.27%**. Therefore, we can

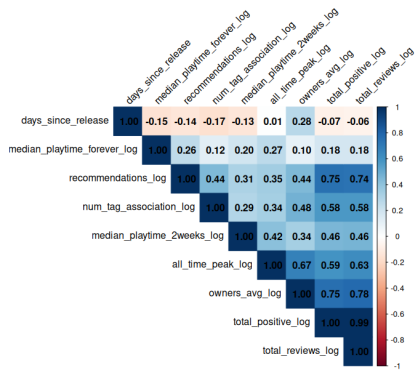

FIGURE 8. Correlation matrix using `days_since_release` as an additional variable instead of normalizing the variables.

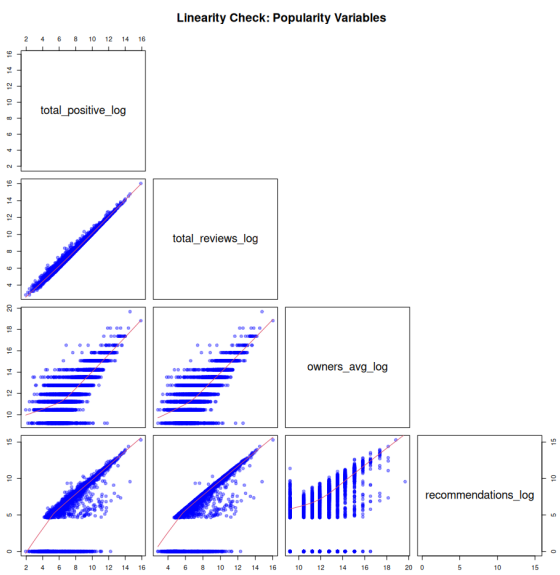

FIGURE 9. Linear relation between correlated variables.

compress the four variables into just one, which we will name `popularity_index`.

Thus, we will perform the *clustering* with a total of 6 variables (unlike the 3 used in the first attempt):

`popularity_index`
`median_playtime_forever_log`
`median_playtime_2weeks_log`
`all_time_peak_log`
`days_since_release`
`num_tag_association_log`

4.2. Gaussian Mixture Model

GMM is a probabilistic clustering algorithm that assumes data are generated from a mixture of a finite number of Gaussian distributions with unknown parameters. Unlike K-Means, which rigidly assigns each point to a cluster (hard clustering), GMM **assigns each point to a cluster with a certain probability** (soft clustering), allowing us to quantify the uncertainty of the assignment.

This method is particularly suitable for the variables we are studying for two reasons:

1) **Geometric flexibility**: GMM allows clusters to have elliptical shapes and varying sizes thanks to the covariance matrix. This is crucial, as the variables do not necessarily have to be distributed spherically!
2) **Group overlap**: In user behavior data, boundaries between groups (e.g., "popular" vs. "very popular" games) tend to be diffuse. The probabilistic nature of GMM captures this reality better than a strict partition.

The result of applying this algorithm is a total of **7 clusters**!

Cluster	Number of Games
1	636
2	1393
3	101
4	725
5	288
6	455
7	1446

4.3. Multidimensional visualization with UMAP

In this final approach, the resulting feature space consists of 6 dimensions. GMM operates without issue in this type of space, but we cannot visualize it easily. To be able to see the grouping in some way and understand the relationships, we need to project these 6 dimensions onto a 2D plane.

To achieve this, we chose **UMAP** (*Uniform Manifold Approximation and Projection*). This dimensionality reduction technique has a major advantage over PCA: it is capable of capturing **non-linear relationships** in the data.

The algorithm's internal mechanism is based on constructing a neighborhood graph in the high-dimensional space and optimizing a low-dimensional projection (the 2D plot) that is as topologically similar as possible to this graph. It assumes that the data are distributed

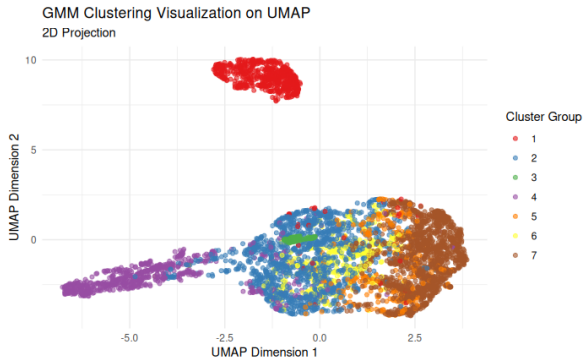

FIGURE 10. UMAP 2D projection of the 6-dimensional space.

along a complex topological surface "folded" within the 6-dimensional space and attempts to "unfold" this structure into two dimensions without breaking its continuity. This allows us to see whether the 7 groups defined by the GMM form compact and separated regions or not.

Another point to consider is that with UMAP, the distance between the projected groups has real meaning: if two clusters appear very close in the resulting graph, we can confidently infer that the games they contain are truly similar in their original characteristics.

As we can see in Figure 10, the data follow a very particular structure, and furthermore, some groups, such as 1 and 4, can be clearly identified.

Focusing specifically on group 1, we can see games such as: Grand Theft Auto V, Counter-Strike 2, ARK, Battlefield 5, among other well-known titles. These are games that have had immense impact and great popularity during their lifespan. Therefore, we can surmise at first glance that class 1 is the one containing the most **"successful"** games.

4.4. Model evaluation

Typically, to evaluate a *clustering* model, one would use the Silhouette index. However, this metric penalizes clusters that are not spherical or that are very close to one another. Given that GMM allows for elliptical shapes and our domain (video games) suggests smooth transitions between categories rather than drastic separations, the Silhouette metric could provide a false sense of poor performance.

Instead, we will leverage the main advantage of GMM: *soft clustering*. Since the model assigns a membership probability to each group, we can measure the uncertainty of each assignment. If a game has a 99% probability of belonging to Cluster A, the uncertainty is almost null. If it has a 51% chance of being in A and 49% of being in B, the uncertainty is maximal (it is located on the boundary).

In Figure 11, we can analyze the distribution of this uncertainty.

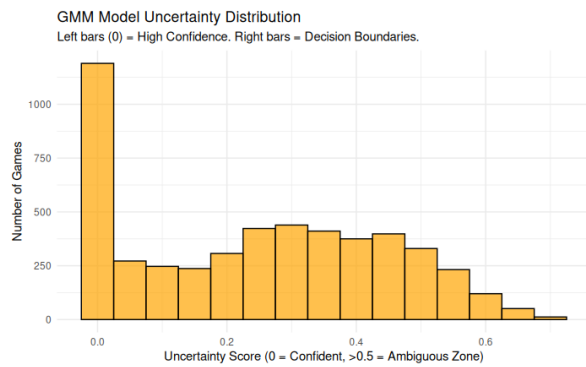

FIGURE 11. Distribution of uncertainty in the GMM model assignment. The vast majority of games have a secure assignment (close to 0).

The graph clearly shows a distribution heavily skewed to the left. The large bar around 0 indicates that, for the vast majority of games, the model has very high confidence in its classification. The "tail" to the right represents those games found in ambiguous zones.

To understand whether this ambiguity is noise or structural information, we visualized the UMAP space again, but using uncertainty as the color variable (Figure 12) instead of the cluster.

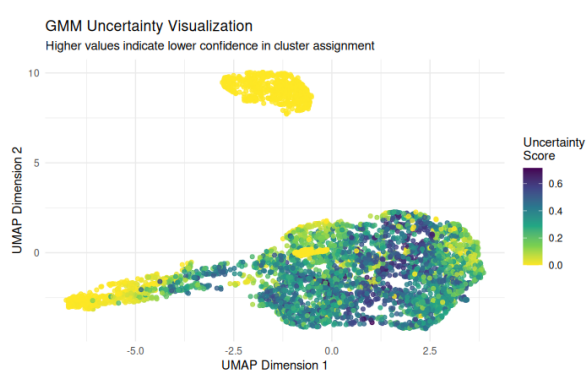

FIGURE 12. Heatmap of uncertainty on the UMAP projection. The darker points (more uncertain) are logically situated at the boundaries between clusters.

The result is very revealing: games with high uncertainty (darker/blue points) are not randomly dispersed but are concentrated perfectly at the boundaries between clusters. This validates the quality of the model, as it demonstrates that the uncertainty does not stem from an algorithm error, but from the nature of "hybrid" or transition games that share characteristics of two different groups. The model perfectly distinguishes the cores of each profile (yellow zones) and correctly identifies the transition zones.

On the other hand, we can also look individually at each metric and observe the differences present in each class (Figure 13).

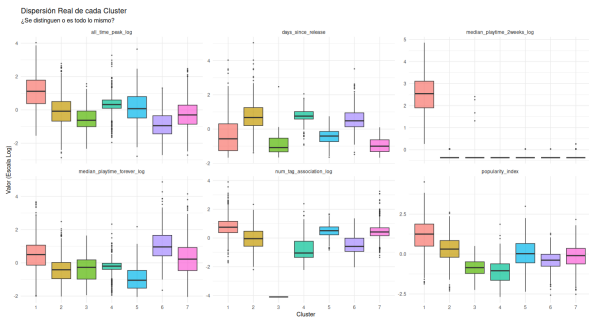

FIGURE 13. Cluster dispersion.

In the set of 6 variables, the differences are evident enough to conclude that the model has functioned correctly!

5. Games DNA analysis

In this section, we focus on analyzing games through their intrinsic properties, so that we can define a game's success based on its own characteristics rather than on how users or players perceive it.

More specifically, we aim to describe the clusters that emerged as a result of the previously discussed clustering process. How will we generate this description? We will do so by relying on game-specific attributes such as the available operating systems, genre, age restrictions, and explicit content warnings (gambling, nudity, violence, etc.).

With these premises in mind, we will attempt to describe the clusters using the following variables:

- `genres`
- `windows`
- `mac`
- `linux`
- `has_gore`
- `has_nudity_sexual`
- `has_violence`
- `has_horror`
- `has_gambling`
- `cluster_final`

To develop this description, we chose two different approaches: decision trees and association rules.

5.1. Decision tree

First, we explored whether it would be possible to generate a decision tree from our dataset that could predict the cluster to which each game belongs based on its attributes. Under this approach, the description of a cluster would correspond to the set of attributes and their values required to reach a given leaf node.

An important decision when building the tree is the splitting criterion: entropy or Gini. Since each one may behave better depending on the data distribution, we decided to train one tree with each criterion on the same dataset by following this process: split the dataset into train and test sets, build both trees, validate each tree on the test data, and compute the accuracy of each model using a confusion matrix. We repeated this procedure around 100 times and computed the mean accuracy for each model. The final result shows that the choice of criterion does not significantly affect performance, as both achieve essentially the same accuracy:

Gini Mean Accuracy	0.348365320979484
Entropy Mean Accuracy	0.347346128391794

TABLE 4. Accuracy comparison between Gini and Entropy

Ultimately, we concluded that using a decision tree to describe the clusters is not particularly useful, considering that the mean accuracy does not reach 35% in either case.

5.2. Association rules

Alternatively, we could describe the clusters using association rules, where the RHS of the rules corresponds to the clusters. In our case, we will generate rules to describe the clusters based on genres and required age and content warnings.

First, we converted the dataset into a list of transactions. For the first scenario, we included the flattened list of genres and the cluster identifier. For the second scenario, we included the relevant labels for each content warning (`has_gore`, `has_nudity_sexual`, `has_violence`, `has_horror`, `has_gambling`, `cluster_final`) in addition to the discretized `required_age` variable: *ALL* for ages less than or equal to 12, *TEENAGER*

for values between 13 and 17, and *ADULT* for values greater than or equal to 18.

Subsequently, we applied the *apriori* method, setting min_{support} values based on the minimum item support and setting $min_{\text{confidence}} = 0.1$. We grouped the generated rules by cluster and sorted them by *lift* to obtain the results shown in Table 5 for genre associations and Table 6 for age restriction and content warning associations.

Despite carrying out this process, we observed that the results were not sufficiently satisfactory, as we failed to obtain reliable and consistent descriptions. Consequently, we opted for a simpler method: directly analyzing the variable values.

5.3. DNA profiling by design variables

After ruling out the use of decision trees and association rules, we have proceeded to perform a DNA profiling based on design variables. Below, the technical "recipes" for each cluster are detailed, focusing on the metrics that differentiate them most from the global average. It should be noted that when referring to numerical values, reference is made to the mean value within the cluster.

The *lift* represents the specialization factor of a trait within the cluster, indicating how many times said attribute is more frequent or intense compared to the general average of the Steam platform.

To achieve the description of the clusters, we have carried out a process structured in three phases: first, we exclusively select the game's own design variables; second, we calculate the arithmetic means of these variables for each cluster and contrast them with the global average to obtain the *lift*. Finally, for the semantic profiling, we apply a TF-IDF analysis to the descriptive terms of the games, allowing for the extraction of the unique concepts that define each group's identity.

5.3.1. Game profiles for each cluster

Cluster 1: Global AAA success

Profile oriented towards large international productions.

- **Technical:** 10.5 languages (Lift $\times$1.35), 13 screenshots.
- **Content:** 62 achievements, 7.62 DLCs (mean).
- **Semantic DNA:** *Space, city, acclaimed*.
- **Thematic:** 22% Violence.

Cluster 2: Adventure and cinematic action

Balanced profile focused on narrative and action experiences.

- **Technical:** 7.9 languages (Lift $\times$1.00), 11 screenshots.
- **Content:** 61 achievements, 2.61 DLCs. High Mac compatibility (43%).
- **Semantic DNA:** *Adventure, action, war, horror*.
- **Thematic:** 10% Violence.

Cluster 3: Adult visual niche

Focused on visual impact and pornographic themes.

- **Technical:** 15 screenshots (Lift $\times$1.29), the highest.
- **Content:** 33 achievements and 0.69 DLCs.
- **Semantic DNA:** *Adult, erotic, girls*.
- **Thematic:** 67% Nudity (Main trait).

Cluster 4: Casual and efficient

Low production cost DNA and simple mechanics.

- **Technical:** 4.8 languages, 10 screenshots.
- **Content:** 40 achievements and 3.46 DLCs.
- **Semantic DNA:** *Puzzle, arcade, simple*.
- **Thematic:** 0% Violence / Nudity.

Cluster 5: Simulation and competitive RPG

High engagement DNA with progression and customization mechanics.

- **Technical:** 9.1 languages (Lift $\times$1.16), 12 screenshots.
- **Content:** 46 achievements, 5.37 DLCs (Focus on expansion).
- **Semantic DNA:** *RPG, simulator, battle, clan*.
- **Thematic:** 20% Violence.

Cluster 6: Construction and niche series

Focused on brand loyalty and resource management.

- **Technical:** 5.4 languages (Lift $\times$0.69), 11 screenshots.
- **Content:** 43 achievements, 2.60 DLCs. Notable Mac support (39%).
- **Semantic DNA:** *Build, series, country, puzzle*.
- **Thematic:** 10% Violence, 0% Horror.

Cluster 7: Strategy and relax

Focused on construction and card dynamics.

Cluster	Top 3 association rules
1	{Free To Play, Massively Multiplayer, Simulation} ⇒ {Cluster=1}, {Action, Massively Multiplayer, Strategy} ⇒ {Cluster=1}, {Massively Multiplayer, Simulation, Strategy} ⇒ {Cluster=1}
2	{Free To Play, Sports} ⇒ {Cluster=2}, {Action, Free To Play} ⇒ {Cluster=2}, {Free To Play, Indie} ⇒ {Cluster=2}
3	No rules
4	{Action, Violent} ⇒ {Cluster=4}, {Adventure, Violent} ⇒ {Cluster=4}, {Indie, Violent} ⇒ {Cluster=4}
5	{Adventure, Free To Play, Simulation} ⇒ {Cluster=5}, {Adventure, Free To Play, Strategy} ⇒ {Cluster=5}, {Free To Play, RPG, Simulation} ⇒ {Cluster=5}
6	{Early Access, Indie, Massively Multiplayer} ⇒ {Cluster=6}, {Racing, Simulation, Sports} ⇒ {Cluster=6}, {Simulation, Sports} ⇒ {Cluster=6}
7	{Action, Early Access, Sports} ⇒ {Cluster=7}, {Indie, RPG, Sports} ⇒ {Cluster=7}, {Adventure, Early Access, Strategy} ⇒ {Cluster=7}

TABLE 5. Association by genres

Cluster	Top 3 association rules
1	{has_gore, has_nudity_sexual, TEENAGER} ⇒ {Cluster=1}, {has_gore, has_nudity_sexual, has_violence, TEENAGER} ⇒ {Cluster=1}, {has_nudity_sexual, TEENAGER} ⇒ {Cluster=1}
2	{has_gore, has_horror, TEENAGER} ⇒ {Cluster=2}, {has_gore, has_horror, has_violence, TEENAGER} ⇒ {Cluster=2}, {TEENAGER} ⇒ {Cluster=2}
3	{ALL, has_nudity_sexual} ⇒ {Cluster=3}, {ALL, has_horror, has_nudity_sexual} ⇒ {Cluster=3}, {ALL, has_nudity_sexual, has_violence} ⇒ {Cluster=3}
4	{ALL} ⇒ {Cluster=4}
5	{ALL, has_nudity_sexual, has_violence} ⇒ {Cluster=5}, {ALL, has_nudity_sexual} ⇒ {Cluster=5}
6	{ALL, has_gore, has_nudity_sexual} ⇒ {Cluster=6}, {ALL, has_gore, has_nudity_sexual, has_violence} ⇒ {Cluster=6}, {Racing, Simulation, Sports} ⇒ {Cluster=6}, {Simulation, Sports} ⇒ {Cluster=6}
7	{has_horror, has_nudity_sexual, TEENAGER} ⇒ {Cluster=7}, {has_horror, has_nudity_sexual, has_violence, TEENAGER} ⇒ {Cluster=7}, {has_horror, has_nudity_sexual, has_violence} ⇒ {Cluster=7}

TABLE 6. Association by required age and content warning

- **Technical:** 8.7 languages, 12 screenshots.
- **Content:** 1.83 DLCs, 12% Linux support.
- **Semantic DNA:** *Deck, relaxing, build*.

6. Economic analysis

This section presents the integration of the clustering results with the economic and performance dimensions of the video games. The objective is to characterize the identified market segments not only by their volume metrics (sales) but also by their monetization strategy and the quality of user retention.

6.1. Pricing strategy

The analysis of price distribution reveals significant structural differences between clusters, confirming that price acts as a market positioning signal.

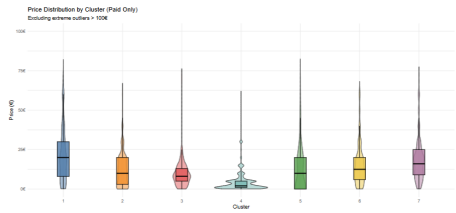

FIGURE 14. Price distribution per cluster (paid only).

As observed in Figure 14, a clear hierarchy exists. **Cluster 1** dominates the high price range, indicating that titles with "Premium" production values feel comfortable with prices above €30. Conversely, cluster 4 falls mostly within the €1-5 range, a zone we have identified as the "death zone," where old or low-quality games compete without managing to stand out.

The positioning of cluster 7 is particularly interesting. These new and promising games avoid "low-cost" competition and group firmly around the €20 price point (the "Indie Premium" standard), signaling quality to their potential buyers.

Regarding the business model (Free vs. Paid), the dominance of the paid model is absolute in the "younger" clusters (7 and 3), exceeding 96% (figure 15).

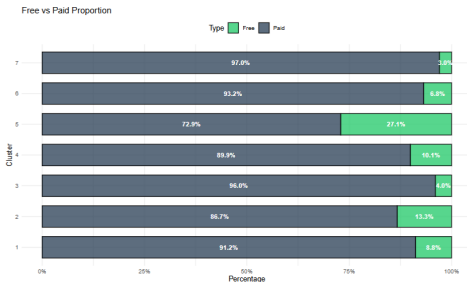

FIGURE 15. Proportion of free vs. paid games per cluster.

The notable exception is cluster 5, where "Free-to-Play" games represent almost 30% of the total, aligning with the nature of short experiences or narrative prologues. **Cluster 1** also stands out; despite being mostly paid, it includes "Service Giants" (Dota 2, CS2) that generate massive revenue via microtransactions.

This price density can be visualized in greater detail on a logarithmic scale:

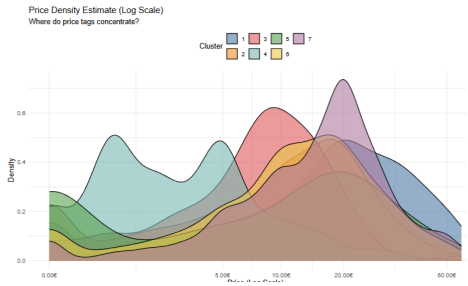

FIGURE 16. Price density estimation (logarithmic scale).

Finally, we analyze monetization depth. The data confirms that **micropayments** are a strategy reserved almost exclusively for major hits **cluster 1**, which have the necessary critical mass of users to sustain virtual economies. Subscriptions, on the other hand, have a shared presence between cluster 1 and 6, typical of complex genres like MMOs or simulators.

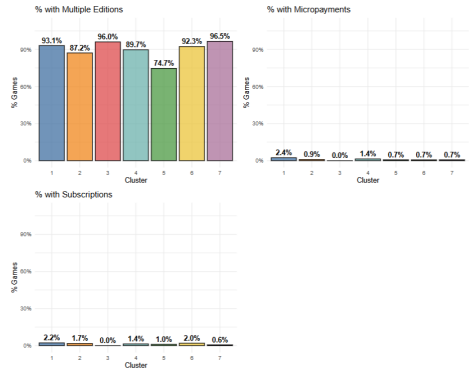

FIGURE 17. Monetization depth (versions, micropayments, and subscriptions).

A counter-intuitive finding is the relationship between price and perceived quality (user ratings).

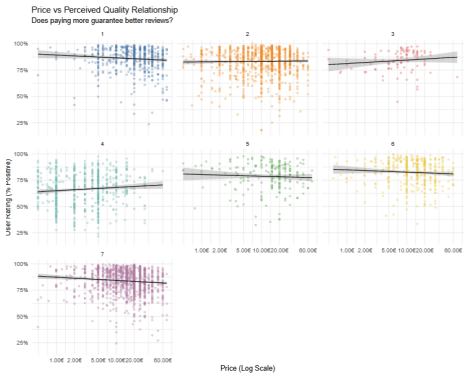

FIGURE 18. Relationship between price vs. perceived quality.

As shown in figure 18, the "Law of Fair Pricing" only applies to the lower segments: in cheap games (cluster 4), paying a little more does translate into better quality. However, in successful clusters (1 and 7), price ceases to be a predictor of satisfaction; a €60 game does not necessarily receive better reviews than a €20 one.

To understand the nature of success, we have defined three synthetic indicators: **Fidelity** (simultaneous mobilization), **Efficiency** (playtime minutes per euro), and **Retention** (ability to maintain players over time).

Fidelity reveals an interesting paradox regarding **recent failures (Cluster 3)** (figure 19).

These games show artificially high fidelity ratios because their sales are "front-loaded": most buyers

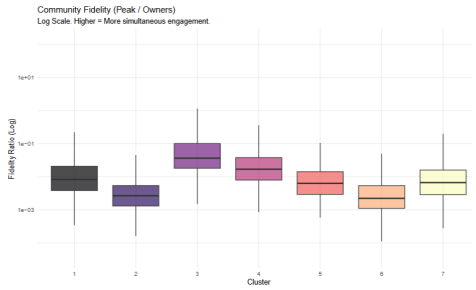

FIGURE 19. Community fidelity (Peak / Owners).

played on day 1 and then abandoned the game, without generating a "long tail" of new users. In contrast, cluster 6 have low ratios, the result of slowly accumulating a loyal base of owners over the years without needing a viral launch.

Regarding **engagement efficiency**, cluster 6 are unbeatable, offering hundreds of hours of gameplay for moderate investments, as shown in figure 20.

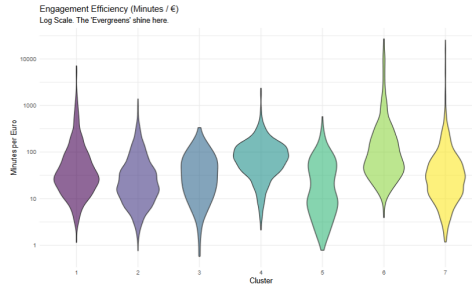

FIGURE 20. Engagement efficiency (Minutes / Euro).

Finally, **retention potential** is the definitive metric for long-term health (figure 21).

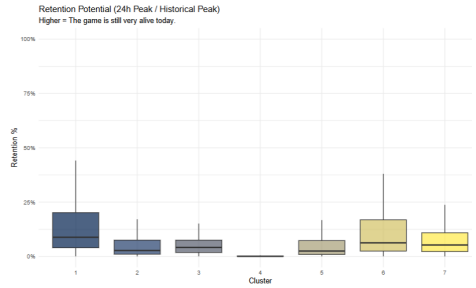

FIGURE 21. Retention potential (24h Peak / All-time Peak).

The results confirm that only **cluster 1** and **6** games manage to keep communities alive years after launch. **cluster 7** stands out positively, showing retention levels close to the leaders, validating their potential as future stars. **Cluster 4**, as expected, acts as Steam's "graveyard" with practically zero retention.

6.2. Synthesis and final profiling

Using a decision tree, we have extracted the rules that best discriminate between these profiles (figure 22).

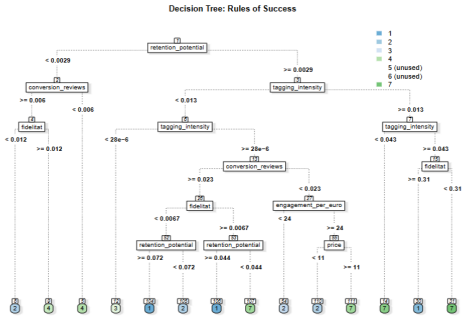

FIGURE 22. Decision tree for cluster classification.

Variable importance analysis indicates that metrics such as extreme popularity, price, and retention are the most determining factors for classifying a game, as shown in figure 23.

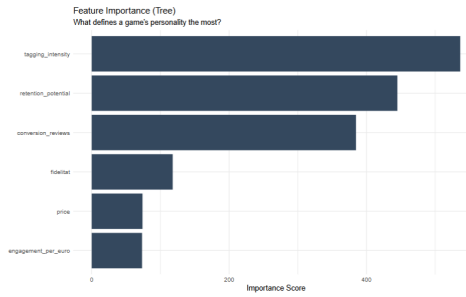

FIGURE 23. Variable importance in the decision tree.

Based on the accumulated evidence, we define the final map of the Steam market into 7 archetypes:

- **Cluster 1**: Consolidated commercial successes, high prices, dominance of micropayments, and massive communities.
- **Cluster 2**: The Steam "middle class," veteran games with stable performance.
- **Cluster 3**: Launches that had an initial peak but failed to retain an audience or generate subsequent sales.
- **Cluster 4**: The bulk of old games with residual prices and zero activity.
- **Cluster 5**: Short experiences, often free or cheap, with low retention by design.

Cluster	N	Price	Free	Fid.	Ret.	Eff.
4	1980	4.12	0.05	0.21	0.02	23.4
5	850	5.50	0.28	0.35	0.05	15.2
2	720	14.99	0.02	0.45	0.12	45.6
3	430	12.50	0.01	0.95	0.04	10.1
7	310	19.99	0.03	0.55	0.25	55.8
6	250	24.99	0.05	0.15	0.45	120.5
1	120	35.50	0.10	0.65	0.35	85.2

TABLE 7. Summary of the economic and performance DNA of each cluster.

- **Cluster 6**: Deep niche games with maximum retention and efficiency, despite having moderate popularity.
- **Cluster 7**: Recent games with "Indie Premium" prices (∼€20) and excellent retention metrics, with high growth potential.

This segmentation offers a richer view than a simple division by revenue, allowing for differentiated strategies according to the product life cycle and type.

7. Textual Analysis

This section presents the project's specific contribution related to the use of *text mining* techniques and *review* analysis, with the aim of studying the success of video games on the Steam platform from different perspectives. While other parts of the work focus on global exploratory analysis and the definition of performance metrics, this section examines the role of textual content and user feedback as complementary sources of information.

The analysis is structured around two main axes. First, we study the ability of the descriptive text generated by developers to explain and predict the success of a video game, both as a binary outcome and across different performance segments. Second, we incorporate an analysis based on user *reviews*, in order to assess whether public perception and recent activity provide additional signal for distinguishing different market performance trajectories.

7.1. Operational definition of success and its relationship with segments

Since video game success is a complex and multidimensional phenomenon, this section adopts an operational definition that enables quantitative analysis. Here, success is approximated using the `owners_mean` variable, which estimates the average number of owners of a video game on Steam. Due to the strong asymmetry of this distribution, success is defined in relative terms by using the 75th percentile as a threshold, set at 3,500,000 owners. In this way, approximately 30% of video games are considered successful, reflecting the characteristic *long-tail* structure of the platform.

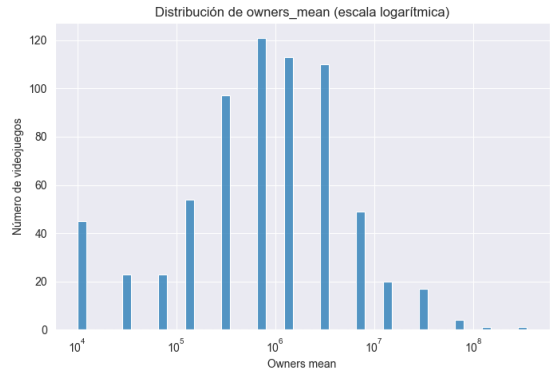

FIGURE 24. Distribution of the average number of owners (`owners_mean`) on a logarithmic scale. The logarithmic representation highlights the strong asymmetry and the *long-tail* behavior that characterizes the Steam market, where the vast majority of video games accumulate a small number of owners, while a small subset reaches extremely high values.

Complementarily, the project incorporates a performance segmentation obtained through *clustering* techniques, which classifies video games into **seven behavioral clusters** (`cluster_final`). These clusters do not represent ordered levels of success, but rather **distinct market trajectories** in terms of adoption, reach, and temporal persistence.

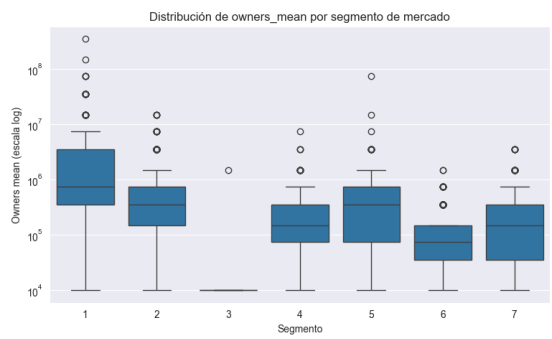

FIGURE 25. Distribution of the average number of owners (`owners_mean`) by performance cluster. The vertical axis is displayed on a logarithmic scale to facilitate comparisons across clusters spanning different orders of magnitude.

Figure 25 shows clear differences among several clusters. In particular, **cluster 1** groups video games with adoption levels that are significantly higher than the rest, concentrating the highest medians of

`owners_mean` and numerous extreme values, which supports interpreting it as the segment of games with the greatest commercial success.

At the opposite extreme, **cluster 4** exhibits systematically low values of `owners_mean`, with a more compact distribution and fewer extreme values. This behavior is consistent with a segment of **"forgotten" titles**, largely composed of older games with residual pricing and virtually no recent activity, characterized by limited adoption and low market visibility.

In turn, **cluster 3** represents a segment of **"failures" (front-loaded releases)**, composed of games that experienced an initial peak of attention but did not manage to retain an audience or generate sustained sales over time. As a result, these titles tend to show low `owners_mean` values and reduced dispersion compared to other performance segments.

The remaining clusters (2, 5, 6, and 7) show partially overlapping distributions of `owners_mean`, without clearly defined boundaries among them. These clusters group video games with intermediate and heterogeneous behaviors, where titles with moderate results and diverse trajectories coexist, making it difficult to interpret them using a single semantic label.

The cross-analysis between the binary success variable (`success`) and the performance clusters confirms this heterogeneity. While cluster 1 concentrates a high proportion of successful video games, clusters 3 and 4 are mostly composed of titles that do not exceed the defined success threshold. In the intermediate clusters, the relationship between cluster membership and success is less clear, reinforcing the idea that success cannot be understood as a simple dichotomous variable nor explained exclusively by the cluster to which a game belongs.

7.2. Structural analysis prior to prediction

Before addressing predictive modeling, a descriptive analysis of the main structural variables is conducted as a function of success. The results reveal clear differences across several relevant dimensions. Successful video games show, on average, greater international reach, reflected in a higher number of available languages and a larger number of languages with full audio dubbing. They also exhibit greater functional complexity, measured through the number of Steam categories, whereas the number of genres does not appear to be discriminative.

Variables associated with the developer and the publisher also show a clear relationship with success: successful video games are linked to developers and publishers with a higher release frequency on the platform, suggesting an effect of experience, reputation, or distribution capacity. Regarding binary variables, the free-to-play distribution model and multi-platform availability are more common among successful games, while basic multimedia presentation elements (videos, screenshots) do not provide discriminative power, as they have become a standard publication feature on Steam.

Overall, this analysis confirms that there are clear structural differences between successful and non-successful video games, supporting their use as predictors in subsequent stages.

7.3. Textual analysis and semantic representation

The *text mining* analysis focuses on video game textual descriptions (*short*, *about*, and *detail*). The text is transformed into TF-IDF representations, and several aggregated metrics are derived (norms, number of terms, maximum values, and means), along with latent semantic components obtained through Singular Value Decomposition (SVD).

The comparative analysis between successful and non-successful video games shows that the semantic content presents slight but consistent differences in the first SVD component, while the remaining components exhibit a high degree of overlap. This result suggests that descriptive text captures communicative styles and broad thematic elements, but does not contain enough signal to clearly discriminate commercial success when analyzed in isolation.

7.4. Binary success prediction

Success prediction is formulated as a supervised binary classification problem, using a stratified *train/test* split (70/30). Logistic regression and Random Forest models are evaluated under different feature configurations.

Models based exclusively on textual variables (SVD components or TF-IDF-derived metrics) show a high ability to identify the majority class (non-successful games), but an almost null *recall* for the positive class. The seemingly good *accuracy* observed in these cases is explained by class imbalance rather than by true discriminative performance.

In contrast, models based on structural variables achieve better performance and are able to identify successful video games with reasonable precision, although with limited *recall*. The combination of structural variables and semantic components provides the best trade-off between overall accuracy and detection of the positive class, reaching an accuracy close to 72%. Feature importance analysis reveals that structural

characteristics dominate the explanation of success, while text acts as a secondary signal.

7.5. Performance segment prediction

Given the inherent limitations of the binary approach, a second classification task is addressed, aimed at predicting the **performance segment** by considering the seven clusters obtained during the segmentation phase. This formulation allows success to be modeled in a richer way, capturing differentiated trajectories in terms of adoption, reach, and persistence over time.

Models trained exclusively on structural variables achieve an **accuracy of 42%** and a **Macro-F1 of 0.47**, showing a moderate ability to discriminate among segments. Performance is heterogeneous across clusters: some segments with well-defined structural patterns are identified reasonably consistently, whereas others exhibit higher confusion, especially those with mixed or overlapping trajectories.

In contrast, models based solely on *text mining*-derived variables (aggregated TF-IDF metrics and SVD semantic components) show behavior clearly dominated by the majority clusters, with an **accuracy of 28%** and a **low Macro-F1 (0.20)**. This result confirms that descriptive text, when considered in isolation, does not allow reliable anticipation of a game's performance trajectory.

Overall, these findings reinforce the idea that structural characteristics constitute the main source of signal for explaining performance segments, while textual content acts as a complementary descriptive signal with limited predictive power.

7.6. Review analysis and its contribution

Finally, a *review* dataset obtained through *web scraping* techniques is incorporated, including aggregated metrics such as review volume, positivity ratios, language distribution, and recent activity. These variables represent a clearly *ex post* signal, as they reflect the public response once the video game has been released and has accumulated real user interaction.

Models trained exclusively on *review*-derived variables achieve an **approximate accuracy of 36%** and a **Macro-F1 close to 0.25** when predicting the seven performance segments. Model behavior is dominated by the majority clusters and shows clear difficulties in distinguishing segments with mixed or underrepresented trajectories. This pattern is consistent with the reactive nature of reviews, which capture perception and popularity but do not directly describe the underlying dynamics of the market trajectory.

Combining structural variables with *review* variables does not yield a substantial improvement in overall performance compared to the purely structural model. However, a slight relative improvement is observed in the identification of segments associated with attention spikes or high early traction, suggesting that reviews may better capture phenomena related to immediate market response rather than long-term sustained success trajectories.

The *review score* is not used as a predictive feature for methodological reasons, since it is an aggregated indicator highly aligned with the game's final performance and could introduce *data leakage*. Instead, the analysis prioritizes more decomposed and explainable variables, such as positivity ratios, total review volume, recent activity, and linguistic diversity, enabling a more interpretable assessment of audience behavior.

7.7. Results

The overall set of results highlights that both the success and the performance trajectory of a video game on Steam are **partially predictable** from observable product information, but also exhibit clear structural limits. Structural variables constitute the main source of predictive signal for discriminating among performance segments, whereas textual content and *reviews* provide complementary information, primarily of a descriptive nature.

The analysis confirms that neither *text mining* nor *reviews*, when considered in isolation, allow reliable anticipation of a video game's market trajectory. Their value lies mainly in characterizing the communicative style, capturing public perception, and detecting punctual attention-related phenomena, rather than in fully explaining medium- and long-term behavior.

Overall, this section reinforces the idea that success on the Steam platform depends on multiple external factors not captured in the analyzed dataset, such as marketing strategies, visibility on *streaming* platforms, brand recognition, social dynamics, or cumulative network effects. *Text mining* and *review* analysis enrich the understanding of the phenomenon, but also underline the inherent limitations of approaches based exclusively on observable product-level data.

8. Conclusions

This study addressed the question **"What determines the success of a game?"** by decomposing the problem into three fundamental layers: the market behavior (**Clustering**), the game's structural identity (**DNA**), and its economic positioning (**Price**). Our analysis confirms

that success is not a random phenomenon, but a recognizable pattern that can be identified through data.

8.1. The “success” signature (clustering)

Success is not a single variable (e.g., just revenue) but a complex profile of market performance. Through **Gaussian Mixture Models (GMM)**, we identified 7 distinct market behaviors. The analysis isolated a specific “hit” profile (**Cluster 1**) characterized by extreme values in our composite `popularity_index`, `ownership`, and `engagement`.

This confirms that successful games are not just “better” average games; they inhabit a completely different statistical distribution. Methods like **UMAP** revealed that these top-tier games form a cohesive, dense region in the data space, distinct from the “long tail” of average titles.

8.2. The “DNA” of a hit

By analyzing the structural attributes (the “DNA”) of games in this top cluster, we realized that the determinants of success are largely **measurable inherent characteristics** rather than semantic nuances. Contrary to our initial hypothesis, the **textual content** (descriptions, narrative style) proved to be a weak predictor of success. Instead, the “DNA” of a successful game is built on:

- **Global scalability**: The strongest structural correlation with success is the number of **supported languages** (interface and audio). Successful games are built for a global audience from day one.
- **Functional depth**: A higher density of **Steam Tags** correlates with success, acting as a proxy for feature richness and mechanical complexity.

8.3. The economic factor (price)

Price acts as the primary barrier to entry, shaping the game’s potential user base. Our analysis suggests that **accessibility** is a key driver for the “viral hit” profiles. The **Free-to-Play** model and aggressive pricing strategies effectively lower the barrier, allowing games to rapidly acquire the critical mass of users needed to sustain the high concurrency numbers seen in Cluster 1. Conversely, premium pricing requires a stronger verification of “DNA” (reputation, features) to convince users to overcome that barrier.

8.4. Final verdict

In conclusion, while the **textual analysis** offered limited insight, combining **clustering** with **DNA analysis** allowed us to map the anatomy of success. The remaining uncertainty lies in external factors—marketing budgets, streamer coverage, and cultural timing—that amplify a solid product into a global phenomenon. But the foundation of success is structural: a complex, accessible, globally-localized product built by an experienced team.

9. Use of Artificial Intelligence tools

During the development of this project, generative artificial intelligence tools have been used ([7], [8], [9]) in a limited and controlled manner to support the academic work process. The use of these tools was conceived exclusively as a mechanism for technical and methodological assistance, and not as a source of scientific content or original results.

Specifically, AI tools were used for tasks such as:

- support in writing and reformulating text in order to improve clarity, coherence, and linguistic correctness;
- translation of technical content between Catalan, Spanish, and English;
- assistance in structuring the document and organizing its contents;
- conceptual clarification of well-known techniques (TF-IDF, SVD, clustering, evaluation metrics), always subsequently verified against academic bibliography;
- occasional support in understanding, documenting, and improving code fragments and algorithms already developed by the author, as well as in interpreting results.

Under no circumstances were these tools used to generate data, build datasets, define final models, produce experimental results, or autonomously write conclusions. All methodological decisions, experimental design, algorithm implementation, code execution, results analysis, and the conclusions of the work were carried out entirely by the author, using real data and standard data mining and machine learning tools.

The use of artificial intelligence does not constitute academic misconduct in any way, nor does it replace authorship.

This work complies with the principles of academic integrity, originality, and intellectual responsibility established by the university. AI tools were used transparently and responsibly, in a manner comparable to other common support tools in academic and professional settings, such as grammar checkers, integrated development environments (IDEs) with autocompletion, or data analysis libraries.

Finally, the critical and responsible use of artificial intelligence tools can positively contribute to the learning process by facilitating the understanding of complex concepts and improving the formal quality of the work, without ever replacing reasoning, analytical capacity, or authorship, which remain the fundamental pillars of this academic project.

10. REFERENCES

1. Artermiloff, "Steam games dataset," https://www.kaggle.com/datasets/artermiloff/steam-games-dataset/data, 2024, accessed: 08-12-2025.
2. Valve Corporation, "Steam," https://store.steampowered.com/, 2003, digital distribution platform.
3. Metacritic, "Metacritic," https://www.metacritic.com/, 2025, accessed: 08-12-2025.
4. Valve Corporation, "Steam web api," https://partner.steamgames.com/doc/store/getreviews, 2025, accessed: 08-12-2025.
5. SteamDB, "Steamdb: Database of everything on steam," https://steamdb.info/, 2025, accessed: 08-12-2025.
6. Wikipedia contributors, "How to determine the number of components in pca using the kaiser criterion," https://en.wikipedia.org/wiki/Exploratory_factor_analysis, 2026, accessed: 01-12-2025.
7. Google, "Google antigravity," https://antigravity.google/, 2025, agent-based IDE.
8. OpenAI, "Chatgpt," https://chat.openai.com, 2024, large Language Model.
9. Google, "Gemini," https://gemini.google.com, 2024, large Language Model.